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Algorithmic Behavior Under Different Spatial Representations in Spatial Computing

Abstract

Spatial computing systems are incredibly capable and are able to capture detailed 3-dimensional representations of spaces; unfortunately algorithms cannot operate on just "raw data". Instead, these spaces must be converted into simplified mathematical representations that computational systems can actually understand. This research investigates how different ways of modeling spaces influences algorithmic behavior and decision making.

This research uses the Apple Vision Pro's spatial mapping capabilities to capture indoor environments and convert them into computational representations. These environments are converted into computational representations such as (but not limited to) point clouds, spatial meshes or graph based structures. Each of these representations is a different way algorithms can go about reasoning an environment. This is pertinent to identifying reachable areas, calculating paths, or modeling how information or agents propagate through space.

This will explore how mathematical representation choice can affect algorithmic outcomes by comparing behaviors across different spatial abstractions while holding the underlying environment constant. Early work involves generating spatial maps and converting them into graph based models suitable for algorithmic analysis, as well as comparing several foundational algorithms including A* and Dijkstra's shortest path algorithms for navigation, as well as breadth first search (BFS) for reachable space detection and propagation through the environment.

By April, this project will implement multiple experiments on evaluating different spatial algorithms and their performance with decision making. The goal in the end is to better understand the computational consequences of simplifying spaces in the real world, while also contributing to the design of more robust spatial computing systems.